

convai

AI SMEs for Physical Work

Real-Time Conversational AI in 3D environments

Purnendu Mukherjee
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Why are we here at AES?

The common thing that unites us is spatial workflows, 3D environments, physical work

Purnendu

Founder and CEO · Convai



AI should experience the world the way humans do, and physics simulations and game engines are the best way to enable that.

The of the primary use cases we see is within enterprise simulations, specifically industrial workflows and other areas where 3D and spatial understanding are critical.

Training is not an event

Soft Skills

Hard Skills



Purely Conversational/ Soft Skills

Resolving conflicts

Sales Force & Customer Education

Leadership and Coaching in Distributed Teams

Environmentally Immersive Soft-Skills

Decision-making under stress

Safety & risk reduction in high-pressure settings

Navigational / Exploratory Experiences

Practicing situational awareness in new sites

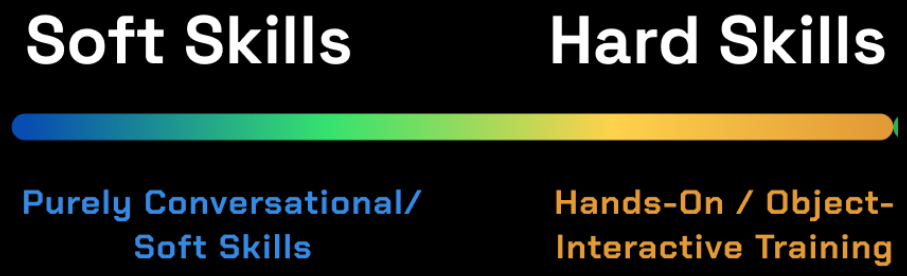
Onboarding and orientation in digital twins

Hands-On / Object-Interactive Training

Complex Product & Equipment Training

Sales demonstrations with hands-on workflows

Safety drills & emergency response practice


Soft Skills
Hard Skills
Purely Conversational/
Soft Skills
Hands-On / Object-
Interactive Training


Soft Skills
Hard Skills
Learning on the Job

AI SME are improving productivity of physical workforce

- A physical workforce learns by doing, with real 3D objects.



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- The AI has to see what the worker sees.



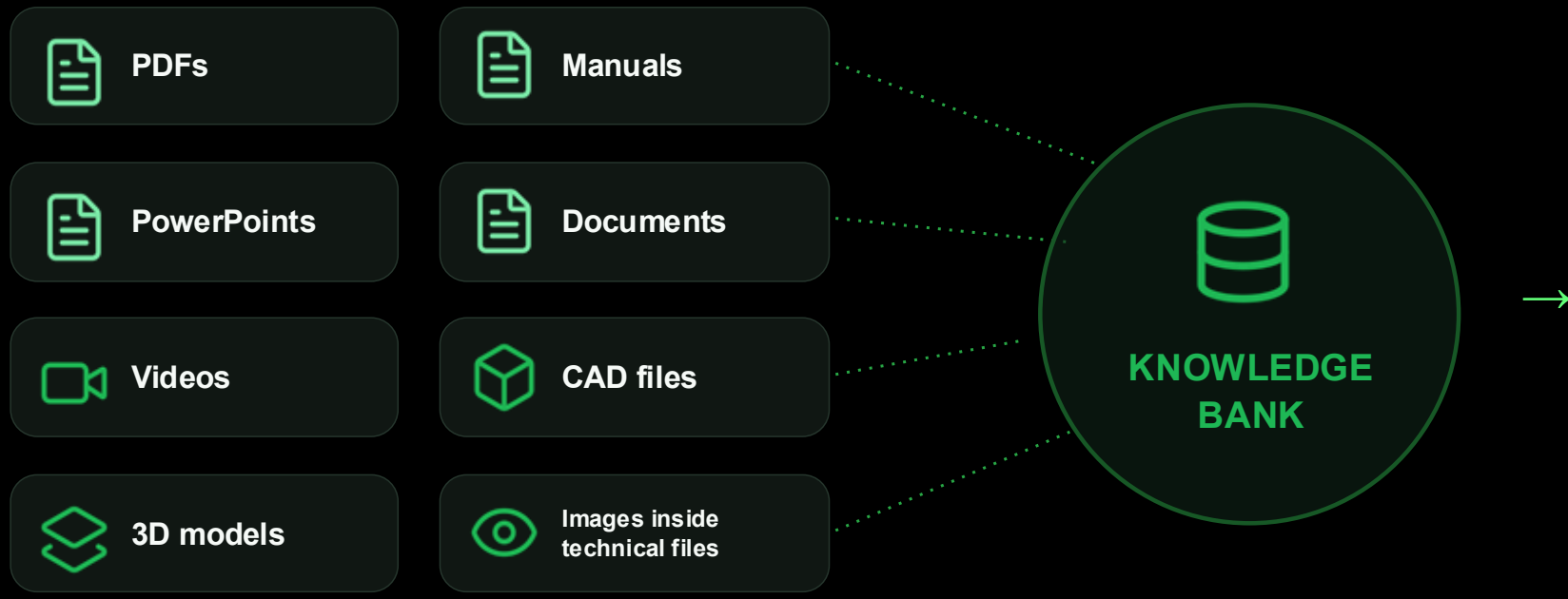
AI SME are improving productivity of physical workforce

- A physical workforce learns by doing, with real 3D objects.
- A chatbot is not enough, and voice alone is not enough.
- The AI has to see what the worker sees.
- It needs full understanding of the machinery's internal architecture, the CAD models, the training manuals and videos, and the rest of the organization's data.



Knowledge From Your Existing Materials

Convai agents can learn from any materials teams already use, including:



→

The goal is to help the agent answer at the level of a subject matter expert

A Day in the Life of a Frontline Worker

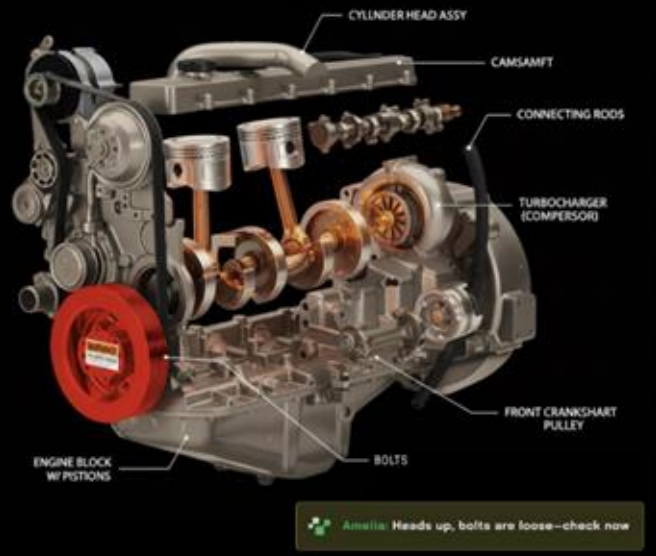
How it works today

- When a worker is stuck on a process or workflow, they ask or call for a subject matter expert.
- Most of these questions are redundant. They are knowledge gaps: finding the right material, guidance and know-how.
- That knowledge sits in a file somewhere, or with a professional who has done the job for 10 years.
- The worker waits, and both people end up waiting or context switching away from their own tasks.



Give Every New Hire an Expert Guide

For complex production environments, Convai can support a smart assistant that helps new hires move through tasks step by step.



In the digital view, the user can inspect the machine, open the 3D model, and understand internal parts



In the real-world view, the user can ask the assistant what to do while standing in front of the physical machine

The assistant can explain procedures, answer questions, show relevant parts, and help users understand what to do next while they are working.

A new employee, three months in

With AI Assistance

- They have had their training. Now they are in front of a machine they need to service or operate.



A new employee, three months in

With AI Assistance

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- They may not know the specifics, or may not fully remember the training.



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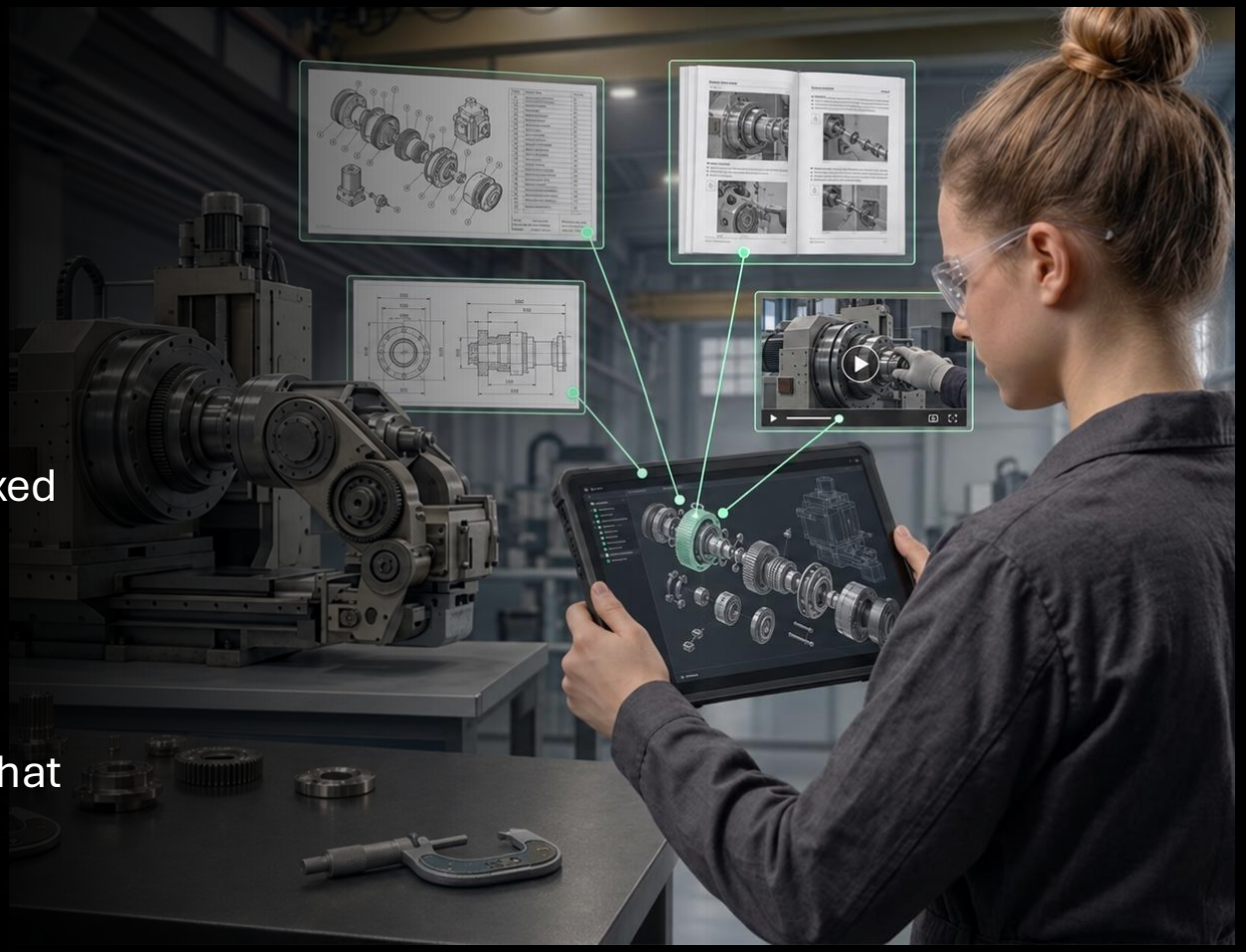
- They have had their training. Now they are in front of a machine they need to service or operate.
- They may not know the specifics, or may not fully remember the training.
- They ask their AI assistant through an AI headset, a mixed reality headset, a phone or a tablet, and it answers.



A new employee, three months in

With AI Assistance

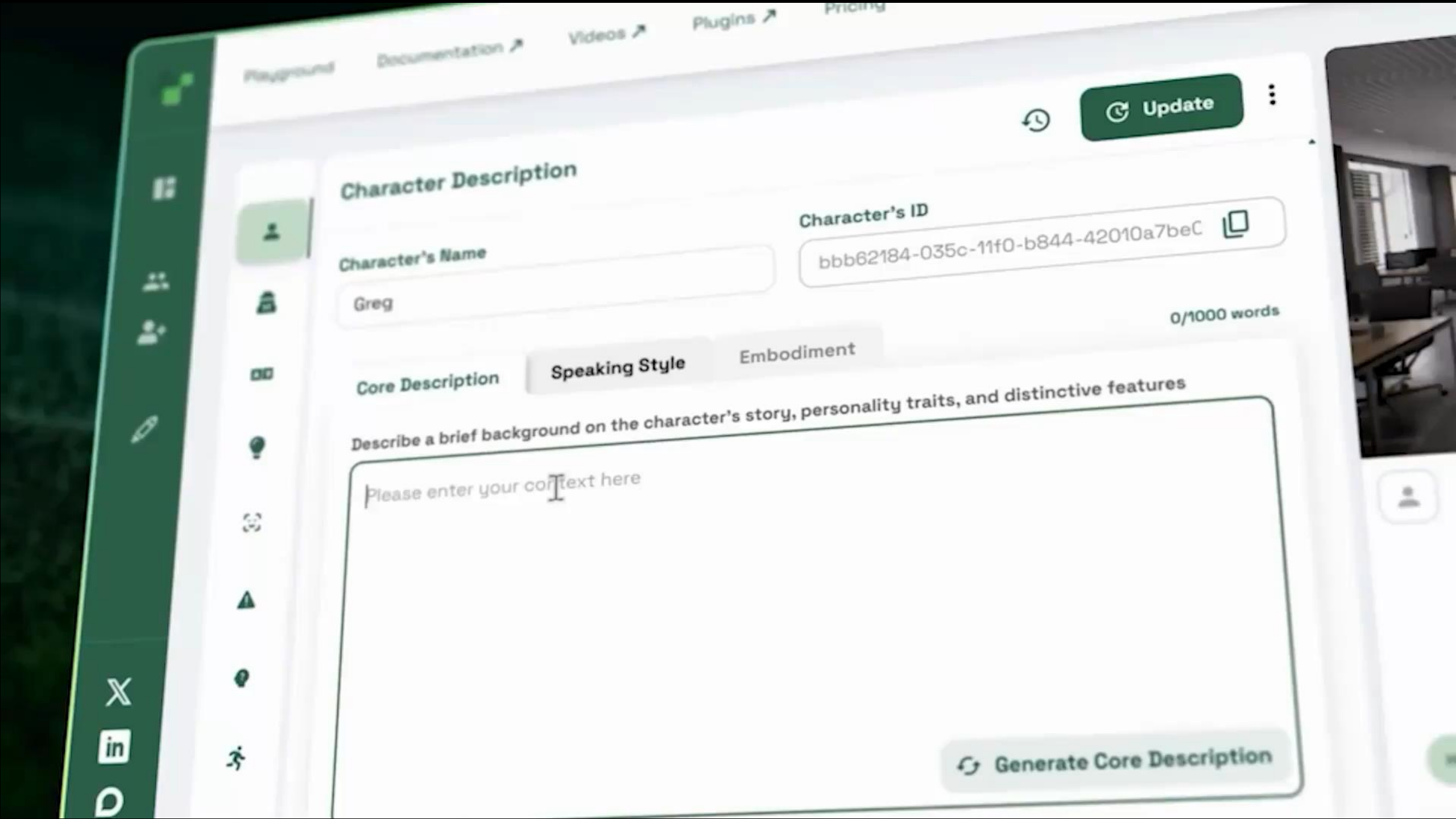
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- They may not know the specifics, or may not fully remember the training.
- They ask their AI assistant through an AI headset, a mixed reality headset, a phone or a tablet, and it answers.
- The AI knows because all the CAD models and related material are connected to it. It becomes an expert on that specific machine or process.





Assembly Studio

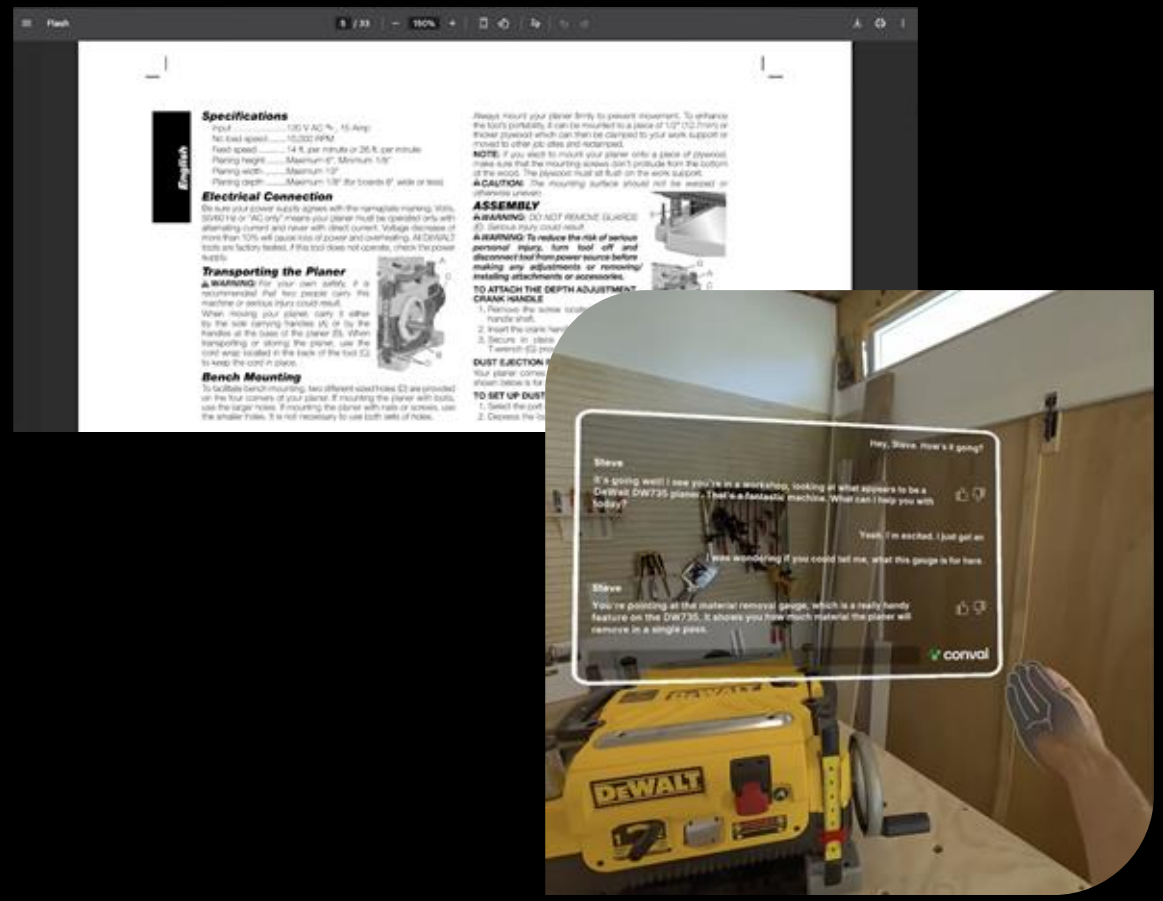
Watch Demo on YouTube


Playground
Documentation ↗
Videos ↗
Plugins ↗
Pricing
Update
Character Description
Character's Name
Greg
Character's ID
bbb62184-035c-11f0-b844-42010a7beC
0/1000 words
Core Description
Speaking Style
Embodiment
Describe a brief background on the character's story, personality traits, and distinctive features
Please enter your context here
Generate Core Description

Built for Complex Tools and Long Procedures

Many production processes involve large tools, long PDF procedures, 3D models, and thousands of individual tasks.

Convai is designed to help turn that information into an interactive assistant that can support employees instead of forcing them to search through long documents or wait for a subject matter expert.



Highly Secure Deployments



Run on Convai Cloud with Enterprise Grade Security

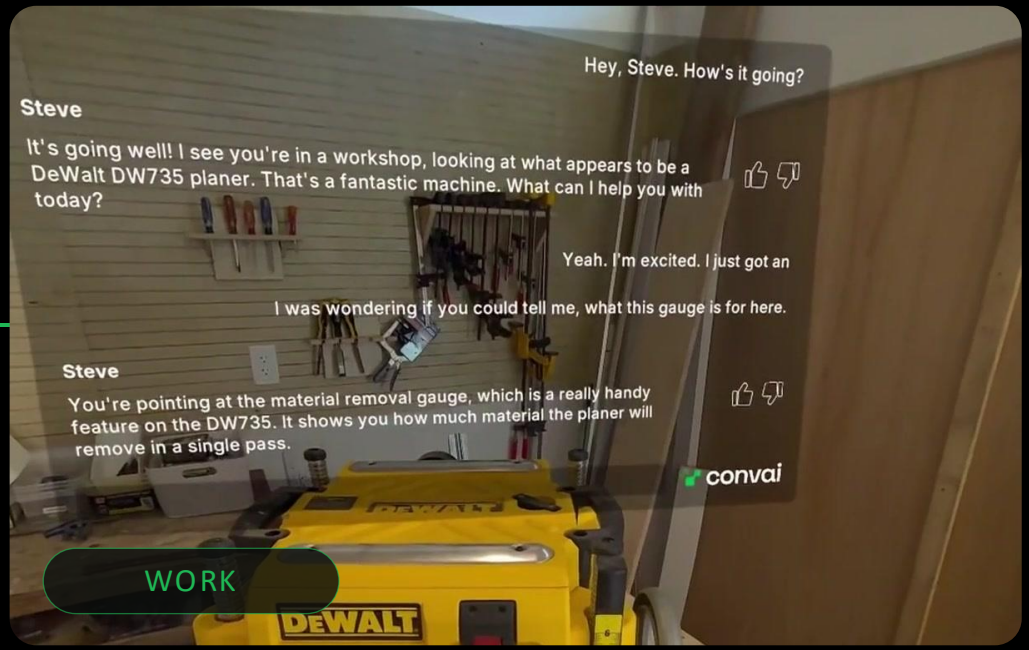


Deploy on your Cloud or Premises for Data Sensitive Workloads

Where workers use it



One mind behind both



Learn on any device

Tablet, phone, laptop or XR headset, fully conversational

Get answers at the real machine

Live video understanding: “What do I do next?”

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Centralized Learning Management

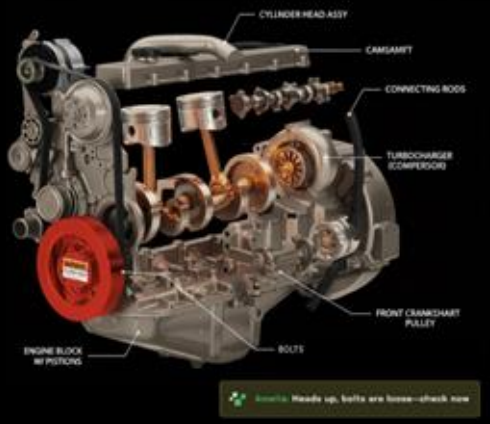
The AI can appear as an avatar, voice assistant, text assistant, or vision-enabled assistant that understands what the user is seeing



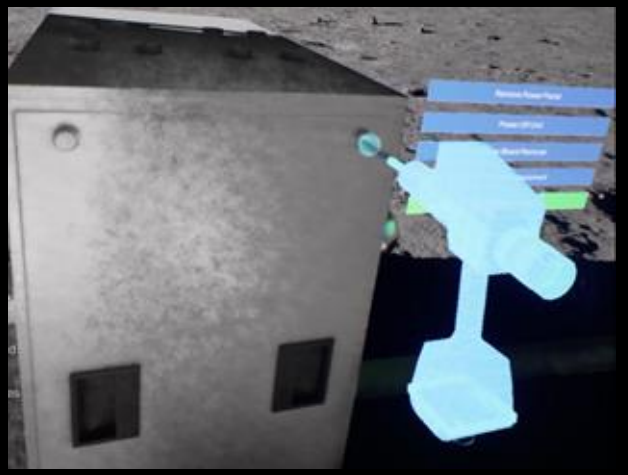
Text-based assistance



Voice-based Handsfree assistance



Voice based AI Assistance when exploring 3D Assets



XR and Simulation based Training in Virtual Environments

One Assistant Across Devices, Wherever the Work Happens

The same agent can be available across multiple devices



Websites



Tablets



Mobile Apps



Augmented Reality

CORE AGENT



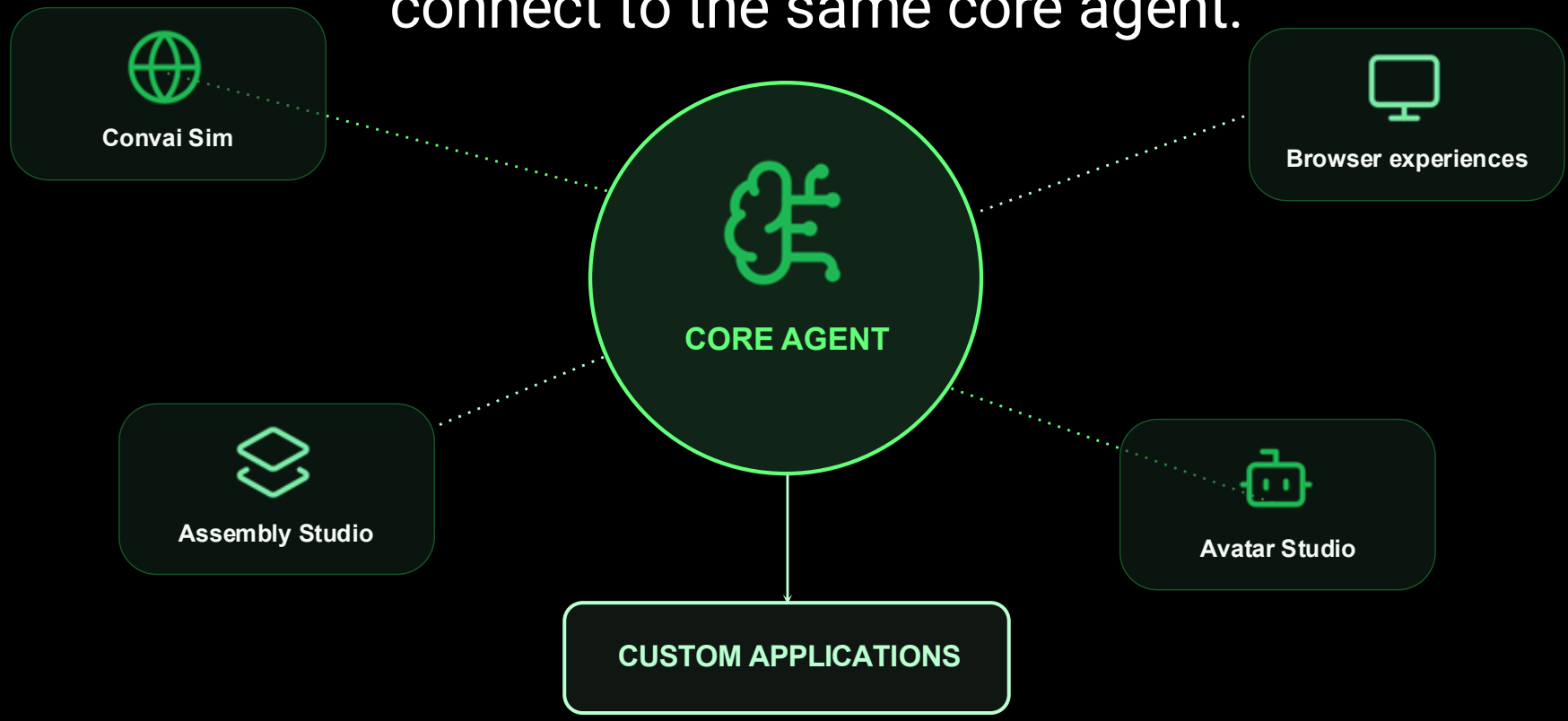
Mixed Reality

This allows a new hire to access help in the format that fits the task

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One Core Platform

Convai Sim, Assembly Studio, Avatar Studio, browser experiences, and custom applications all connect to the same core agent.



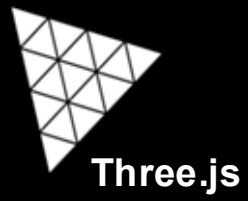
Teams can choose the experience they need, while keeping the same underlying AI, knowledge, and data connections.

Integrates with any Training Simulation Software

with Native Plugins available for









Compliant with Enterprise Requirements

Adhering to the highest standards for information security management, ensuring your data is secure.



ISO/IEC 27001:2022 — Certified

Convai's Information Security Management System is independently certified to the internationally recognized standard for managing information-security risks and controls.



SOC 2 — Compliant

Convai's security controls have been evaluated against applicable AICPA Trust Services Criteria, supporting the secure and responsible handling of customer systems and data.

[Learn more about Enterprise Tier Benefits]

Let’s have a chat

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Let's connect

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@purn3ndu

Thank You


Narrative Driven
Role Playing
Realistic Voices
Accurate Knowledge
Vision Perception
Long Term Memory
Complex Actions
Embodied Animations
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